

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(d)	(i)		$I = \frac{3}{12} = 0.25 \text{ [A] (1)}$ $\text{Rearrangement: } R = \frac{12}{0.25 \text{ ecf}} (1)$ $= 48 \text{ } [\Omega] (1)$ Alternative: $I = \frac{3}{12} = 0.25 \text{ [A] (1)}$ $\text{Rearrangement: } R = \frac{P}{I^2} = \frac{3}{0.25^2 \text{ ecf}}$ $= 48 \text{ } [\Omega] (1)$		3		3	3	3
		(ii)		[Total resistance decreases] because the {lamp / thermistor} is now in parallel or there is another path for the current (1) So the parallel resistance is lower [than the thermistor resistance on its own so the total resistance is lower] (1)	1	1		2		2
				Question 7 total	3	9	2	14	8	14